

Chromatin architecture of clustered TF-paralog loci: CTCF insulation, ATAC accessibility, and ENCODE-rE2G enhancer wiring ohnolog vs. small-scale-duplication (tandem) clusters

Iteration 19 — Paralogous TF project
ETS P01 genomics platform

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Abstract

We profile three complementary layers of regulatory architecture over the *same* 77 clustered TF-paralog loci (47 **clustered cis-ohnolog**, 30 **clustered SSD-tandem**) from the iteration-16 four-group duplication catalog: **CTCF** (insulation), **ATAC-seq** (accessibility) and the new **ENCODE-rE2G** (enhancer→gene wiring). *CTCF* (ENCODE GM12878 ENCFF796WRU and CD4 T cell ENCFF858TLX): bulk insulator density is indistinguishable between groups (all $p > 0.7$), and the only separating metric — more CTCF peaks *between* the two ohnolog members — is explained by their larger inter-member spacing. *ATAC* and *E2G* (regulatory CD4 T cell / Treg, matching the project’s autoimmunity focus; ENCFF519QMA and ENCFF393GIF) reveal what the CTCF layer does not: the two groups differ sharply in accessibility and enhancer wiring. **clustered SSD-tandem** paralogs live in a shared, highly accessible regulatory neighbourhood, whereas **clustered cis-ohnolog** paralogs — though also clustered — have sparser, more independent wiring. ATAC promoters of SSD-tandem members sit essentially on top of an accessible peak (median nearest peak 446 bp vs. 25,492 bp for ohnolog; Mann–Whitney U $p = 1.8 \times 10^{-4}$), and E2G predicts that SSD-tandem paralogs *share the same enhancers*: a median of 10.5 enhancer elements are linked to *both* members (vs. 0 for ohnolog; $p = 3.3 \times 10^{-5}$), and 63% of SSD-tandem pairs share ≥ 1 enhancer vs. 19% of ohnolog pairs. The shared-enhancer signature reproduces in *all six* lymphoid cell types tested (Treg plus five additional ATAC-based ENCODE-rE2G sets: memory CD4, CD8, Jurkat, NK and B cells; ohnolog median = 0 in every one; p from 1.7×10^{-5} to 4.4×10^{-3}).

1 Data and provenance

Duplication catalog (reused, not recomputed). The two clustered classes and their locus coordinates were taken verbatim from iteration 16 (TF_dup_2x2_classification.tsv, clustered_paralogs.noKRAB hg38 / GENCODE v45). Analysis is restricted to the **clustered cis-ohnolog** ($n = 47$ loci) and **clustered SSD-tandem** ($n = 30$ loci) classes; the single chrY locus (HSFY1/HSFY2) is excluded for consistency with the CTCF baseline. Windows = cluster span ± 100 kb, clipped to hg38 chromosome bounds (identical to the CTCF analysis; clustered_loci_windows.pm100kb.tsv).

(A) CTCF insulation. ENCODE CTCF ChIP-seq, two cell types: ENCSR000DZN / file ENCFF796WRU (GM12878 B-lymphoblastoid, conservative-IDR narrowPeak, 44,217 peaks; constitutive reference) and ENCSR470KCE / file ENCFF858TLX (primary resting CD4 $\alpha\beta$ T cell, IDR narrowPeak, 40,346 peaks; matched lineage for the Treg accessibility data). Both GRCh38, downloaded 2026-07-18.

(B) ATAC-seq accessibility. ENCODE experiment ENCSR159GFS (ATAC-seq, biosample CD4⁺CD25⁺ $\alpha\beta$ regulatory T cell / Treg), file ENCFF519QMA — *IDR-thresholded peaks*, bed narrowPeak, GRCh38; 57,636 peaks. Fed to the CTCF pipeline scripts (03/04/05/06) via `--subdir atac`, so the binning, per-locus, flank and paired-promoter code is reused unchanged (in that subdir “peaks” denotes ATAC accessibility peaks).

(C) ENCODE-rE2G enhancer–gene links (the “new E2G dataset”). Annotation ENCSR7590VP (“enhancer gene links in regulatory CD4 T cells”; multiomic ENCODE-rE2G model combining ATAC-seq, H3K27ac, Hi-C and RNA), file ENCFF393GIF — *thresholded element–gene links*, bed3+, GRCh38; 295,377 links / 93,685 unique regulatory elements. Each row is a regulatory element (chr,start,end) predicted to regulate a **TargetGene**, with an ABC/rE2G Score, DistanceToTSS, per-element ATAC accessibility (**Open**) and a class label (promoter / lncRNA / intergenic / genic). Both files downloaded 2026-07-18 from encodeproject.org (provenance in log/encode_download.log).

2 Methods

CTCF (insulation). Each clustered locus, extended by ± 100 kb, was profiled with scripts 03/04/05/06: per-locus CTCF peaks/kb in the window / body / flanks; observed/expected vs. the genome-wide CTCF rate; nearest CTCF peak to each member TSS; and the number of CTCF peaks lying *between* the two paralog members (both raw and normalized to inter-member distance). Run for both the GM12878 and CD4 T-cell peak sets. **ATAC (accessibility mirror).** Each locus window was profiled exactly as for CTCF: per-locus ATAC peaks/kb in the window, locus body and flanks; observed/expected vs. the genome-wide ATAC rate; a length-normalized metagene (20 fixed flank bins + 20 scaled body bins per side); distance from each member TSS to the nearest ATAC peak; and peaks between the two promoters. **E2G (script 07_e2g_promoters.py).** (1) *Per-locus element density*: unique elements (deduped by coordinate) counted in the window/body/flanks, elements/kb and obs/exp vs. a genome-wide baseline of 0.031 elements/kb. (2) *Gene-pair promoter wiring*: for each cluster we collected all E2G links whose **TargetGene** is a cluster member, and counted per member the distinct enhancer elements landing on that promoter; per pair we counted enhancer elements linked to *both* paralogs (“shared enhancers”), both genome-wide and restricted to the ± 100 kb window. **Statistics.** All two-group tests are two-sided **Mann–Whitney U**, **clustered cis-ohnolog** vs. **clustered SSD-tandem**, reported with U , p , per-group n , medians and rank-biserial effect size (results/ctcf/atac/ctcf_group_comparison.tsv, results/e2g/e2g_group_comparison.tsv).

3 Results

3.1 Integrated view: three assays over one locus set

Because all three assays were run over the identical set of clustered loci, their headline metrics can be read side by side (Fig. 1, Table 1). They separate the two groups in a mutually consistent way. **CTCF** places *more* insulator peaks between the two ohnolog members (median 3 vs. 0 for SSD-tandem; $p = 8.9 \times 10^{-3}$) — but, as shown below, this is a consequence of ohnolog members sitting farther apart, not of denser intervening insulation. **ATAC** shows SSD-tandem promoters essentially on top of an open-chromatin peak (nearest ATAC peak 446 bp vs. 25,492 bp; $p = 1.8 \times 10^{-4}$). **E2G** shows SSD-tandem paralogs sharing a median of 10.5 enhancers versus 0 for ohnolog ($p = 3.3 \times 10^{-5}$). The three layers agree on a single axis: ohnolog paralog pairs are wired

for regulatory *independence*, SSD-tandem pairs for *co-regulation*.

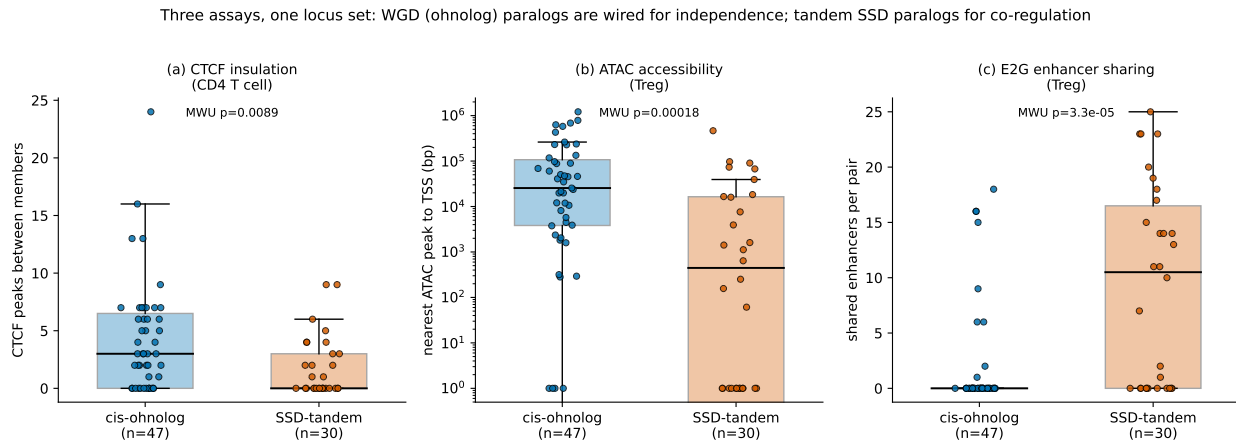


Figure 1: Three assays, one locus set (47 **clustered cis-ohnolog** blue, 30 **clustered SSD-tandem** orange). (a) CTCF peaks between the two paralog members (CD4 T cell); (b) distance from a member TSS to the nearest ATAC peak (Treg; log axis); (c) enhancer elements shared by both paralogs (Treg E2G). Boxes = median/IQR, points = loci, p = Mann–Whitney U.

Table 1: Cross-assay summary over the same 77 clustered loci (Mann–Whitney U, **clustered cis-ohnolog** vs. **clustered SSD-tandem**; results/integrated_cross_assay_summary.tsv).

Assay (layer)	Headline metric	median (ohno)	median (SSD)	p
CTCF — insulation (CD4 T)	CTCF peaks between the two members	3	0	8.9×10^{-3}
ATAC — accessibility (Treg)	Nearest ATAC peak to member TSS (bp)	25,492	446	1.8×10^{-4}
E2G — wiring (Treg)	Enhancers shared by both paralogs	0	10.5	3.3×10^{-5}

3.2 CTCF insulation: bulk density is equal; only between-member counts differ

Both clustered groups are CTCF-enriched over the genome-wide expectation (observed/expected ~ 1.5 – $2.1\times$), and bulk CTCF density across the ± 100 kb window is statistically indistinguishable between them in both cell types (Table 2; all $p > 0.7$ for peaks/kb window, body and flanks). The one metric that separates the groups is the *number of CTCF peaks between the two paralog members*: median 3 (ohnolog) vs. 1 (SSD) in GM12878 ($p = 3.5 \times 10^{-3}$) and 3 vs. 0 in CD4 T cells ($p = 8.9 \times 10^{-3}$). However, ohnolog members are also farther apart (median inter-member TSS distance 74 kb vs. 49 kb), and once the between-member CTCF count is normalized to that distance the difference is no longer significant ($p = 0.27$ GM12878, $p = 0.067$ CD4). In other words, ohnolog pairs are separated by *more* insulator elements because they span *more* DNA, not because the intervening chromatin is intrinsically denser in CTCF. CTCF thus does not, on its own, distinguish the two duplication modes — the distinction emerges only in the accessibility and enhancer-wiring layers below.

Table 2: CTCF architecture, **clustered cis-ohnolog** vs. **clustered SSD-tandem** (Mann–Whitney U, two-sided; $n = 47$ vs. 30 loci), in two cell types. “between members” = CTCF peaks strictly between the outermost member TSSs.

Metric	GM12878 (constitutive)			CD4 T cell (matched)		
	med (ohno)	med (SSD)	p	med (ohno)	med (SSD)	p
CTCF peaks/kb (window)	0.025	0.022	0.92	0.026	0.027	0.89
Observed/expected CTCF (window)	1.70	1.52	0.92	1.96	2.05	0.89
CTCF peaks between members	3.0	1.0	3.5×10^{-3}	3.0	0.0	8.9×10^{-3}
...normalized to distance	0.023	0.018	0.27	0.019	0.0	0.067

3.3 ATAC accessibility: SSD-tandem promoters are more accessible

Across the ± 100 kb window, **clustered SSD-tandem** loci are more accessible than **clustered cis-ohnolog** loci (observed/expected ATAC 2.58 vs. 1.19; Mann–Whitney U $p = 0.016$), with the effect concentrated in the locus body (ATAC peaks/kb 0.067 vs. 0.019; $p = 0.010$). The sharpest separation is at the promoters themselves: the mean distance from a member TSS to its nearest ATAC peak is 446 bp for SSD-tandem members versus 25,492 bp for ohnolog members (Mann–Whitney U $p = 1.8 \times 10^{-4}$, rank-biserial -0.51 ; Table 3). In other words, SSD-tandem paralog promoters sit essentially on top of an open-chromatin peak, while ohnolog paralog promoters are, on average, tens of kb from the nearest ATAC peak. The number of ATAC peaks lying *between* the two promoters does not differ ($p = 0.68$).

Table 3: ATAC-seq accessibility, **clustered cis-ohnolog** vs. **clustered SSD-tandem** (Treg, ENCF519QMA; Mann–Whitney U, two-sided; $n = 47$ vs. 30 loci).

Metric	median (ohnolog)	median (SSD)	p	rank-biserial
ATAC peaks/kb (window)	0.022	0.048	0.016	+0.33
ATAC peaks/kb (locus body)	0.019	0.067	0.010	+0.35
Observed/expected ATAC (window)	1.19	2.58	0.016	+0.33
Nearest ATAC peak to member TSS (bp, mean)	25,492	446	1.8×10^{-4}	-0.51
ATAC peaks between the two promoters	2.0	2.5	0.68	+0.06

3.4 E2G: SSD-tandem paralogs share enhancers; ohnologs do not

The ENCODE-rE2G enhancer→gene links reveal the wiring behind the accessibility difference (Table 4). **clustered SSD-tandem** pairs receive denser regulatory input on their promoters (median 32.5 links vs. 9; $p = 2.7 \times 10^{-4}$; 22.5 vs. 9 distinct enhancer elements; $p = 1.0 \times 10^{-3}$). The most striking difference is *enhancer sharing*: a median of 10.5 enhancer elements are predicted to regulate *both* SSD-tandem paralogs, versus 0 for ohnolog pairs (Mann–Whitney U $p = 3.3 \times 10^{-5}$, rank-biserial $+0.49$; Fig. 2b). Expressed as a fraction of the pair’s total enhancer complement, half of the enhancers of an SSD-tandem pair are shared (shared fraction median 0.50 vs. 0; $p = 1.2 \times 10^{-3}$), and 63% of SSD-tandem pairs share at least one enhancer versus 19% of ohnolog pairs. Restricting to enhancers physically inside the ± 100 kb window preserves the effect (median 6 vs. 0 shared; $p = 7.7 \times 10^{-5}$). Per-element accessibility (**Open**) is similar between groups ($p = 0.35$), so the difference is in *how many* elements are shared, not in how open each one is.

Table 4: ENCODE-rE2G enhancer–gene wiring, **clustered cis-ohnolog** vs. **clustered SSD-tandem** (Treg, ENCF393GIF; Mann–Whitney U, two-sided).

Metric	median (ohnolog)	median (SSD)	p	rank-biserial
E2G elements/kb (window)	0.030	0.066	5.5×10^{-3}	+0.38
E2G observed/expected (window)	0.97	2.14	5.5×10^{-3}	+0.38
E2G links to pair promoters (total)	9.0	32.5	2.7×10^{-4}	+0.49
Distinct enhancer elements on promoters	9.0	22.5	1.0×10^{-5}	+0.44
Shared enhancers (both paralogs)	0.0	10.5	3.3×10^{-5}	+0.49
Shared enhancers within ± 100 kb	0.0	6.0	7.7×10^{-5}	+0.46
Shared-enhancer fraction	0.0	0.50	1.2×10^{-3}	+0.51
Mean enhancer accessibility (Open)	9.37	10.00	0.35	+0.16

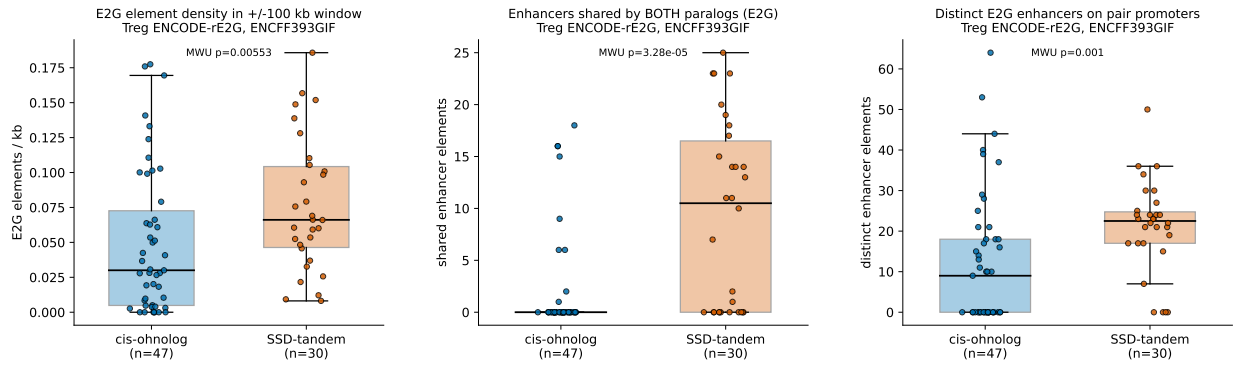


Figure 2: ENCODE-rE2G, **clustered cis-ohnolog** (blue) vs. **clustered SSD-tandem** (orange). (a) Per-locus E2G element density in the ± 100 kb window. (b) Enhancer elements shared by *both* paralogs of a pair — the headline result. (c) Distinct E2G enhancer elements landing on the pair’s promoters. Boxes show the median and IQR; points are individual loci; p -values are Mann–Whitney U.



Figure 3: ATAC-seq (Treg, ENCF519QMA). Left: per-locus ATAC accessibility (peaks/kb) by group. Right: paired-promoter ATAC summary. (Figures generated by the reused CTCF scripts under the `atac` subdir; axis label “CTCF” in the shared script denotes ATAC peaks here.)

3.5 Reproducibility across a 6-cell-type lymphoid panel

To test whether the enhancer-sharing signature is specific to Treg, we re-ran the E2G analysis on five additional ATAC-based ENCODE-rE2G sets (same extended model, hg38). ENCODE’s ATAC-based rE2G resource contains only three T-cell subtypes beyond Treg (memory CD4 ENCF440AKC, CD8 ENCF993UOE, and the T-ALL line Jurkat ENCF705FWU); we added natural-killer (ENCF359ZSL) and B cells (ENCF186ABX) as lymphoid controls. The result reproduces in *every* cell type (Table 5, Fig. 4): the median number of enhancers shared by both paralogs is **0 for ohnolog pairs in all six cell types** versus 6–10.5 for SSD-tandem pairs, with Mann–Whitney U

p ranging from 1.7×10^{-5} (B cell) to 4.4×10^{-3} (Jurkat, the leukemic line, is the weakest but still significant). The per-locus obs/exp E2G density difference (SSD > ohnolog) likewise reproduces in all non-leukemic sets ($p \leq 0.03$). The signature is therefore a robust property of the paralog loci, not an artifact of one cell type.

Table 5: Reproducibility of the shared-enhancer result across six lymphoid cell types (ATAC-based ENCODE-rE2G; Mann–Whitney U, two-sided, **clustered cis-ohnolog** vs. **clustered SSD-tandem**; medians = shared enhancers per pair). Bona-fide T cells in the top block; lymphoid controls below.

Cell type	file	median (ohnolog)	median (SSD)	p	rank-biserial
Treg	ENCFF393GIF	0	10.5	3.3×10^{-5}	+0.49
memory CD4 T	ENCFF440AKC	0	7.5	3.9×10^{-5}	+0.47
CD8 T	ENCFF993UOE	0	10.5	2.5×10^{-4}	+0.43
Jurkat (T-ALL)	ENCFF705FWU	0	7.0	4.4×10^{-3}	+0.35
NK (control)	ENCFF359ZSL	0	6.0	1.1×10^{-3}	+0.38
B cell (control)	ENCFF186ABX	0	9.5	1.7×10^{-5}	+0.51

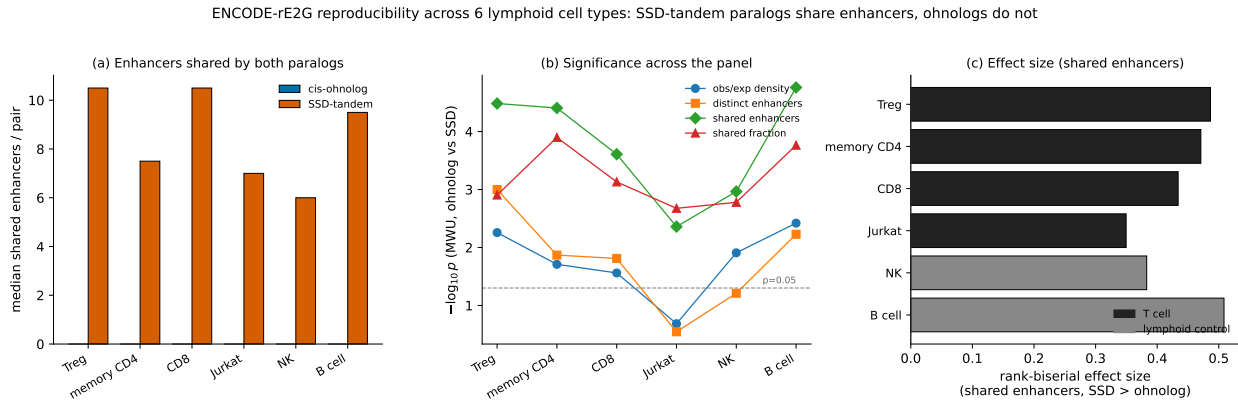


Figure 4: ENCODE-rE2G reproducibility across six lymphoid cell types. (a) Median enhancers shared by both paralogs, **clustered cis-ohnolog** (blue) vs. **clustered SSD-tandem** (orange) — ohnolog is 0 everywhere. (b) $-\log_{10} p$ (Mann–Whitney U) for four E2G metrics across the panel; dashed line $p = 0.05$. (c) Rank-biserial effect size for the shared-enhancer test; dark bars = T cells, grey = lymphoid controls.

4 Interpretation

The three assays layer into one coherent picture. The two clustered groups do *not* differ in their bulk insulator (CTCF) density — CTCF alone cannot tell a WGD cluster from an SSD tandem cluster, and the apparent between-member difference is a spacing artifact. The distinction lives entirely in the accessibility and wiring layers. Tandem SSD paralogs — recently duplicated and physically adjacent — occupy a single, highly accessible regulatory neighbourhood and are co-regulated by a shared enhancer repertoire; roughly half of a tandem pair’s enhancers act on both genes. Cis-ohnolog paralogs, despite also being “clustered”, retain sparser and more *independent* enhancer wiring, with promoters that are frequently far from the nearest accessible peak, and (being farther apart) are separated by more insulator elements. Together the picture is that WGD-derived clustered paralogs are wired for regulatory *independence* (insulated, sparse, unshared enhancers), whereas SSD tandem paralogs are wired for *co-regulation* (open, shared enhancers). For

the ETS/P01 dosage-buffering thesis this is directly relevant: shared-enhancer co-regulation predicts *correlated* dosage between tandem paralogs (a lesion in the shared enhancer would move both), while independent wiring predicts that ohnolog paralogs can be dialed separately — the substrate for paralog buffering.

5 Caveats

(i) Accessibility and enhancer-gene wiring are cell-type specific (Treg here, with CTCF from GM12878 and resting CD4 T cells), unlike largely constitutive CTCF. (ii) ENCODE-rE2G links are model predictions (multiomic ABC-style), so “shared enhancer” counts depend on the score threshold baked into the released file. (iii) The two groups are small ($n = 47$ vs. 30 loci); tests are reported at the locus level. (iv) Some cluster members are readthrough / non-standard symbols that do not match a **TargetGene** and therefore contribute zero links — a conservative bias against detecting wiring. (v) Extended windows can overlap for nearby clusters and were clipped to chromosome bounds.

Files. CTCF: `results/ctcf/` (GM12878), `results/ctcf/cd4/`; ATAC: `results/ctcf/atac/`; E2G: `results/e2g/` (+ per-cell-type subdirs); integration: `results/integrated_cross_assay_summary.tsv`, `results/figures/integrated_th` figures under `results/figures/{atac,e2g}/`; scripts 03--09; logs in `log/`.